

TrES-3b

R-1083

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160513 · created 2026-07-30T15:55:01+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457521.9087 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.013
Transit depth (Rp/Rs)^2	3.86 +/- 0.51 [%]
Orbital Inclination (inc)	82.24 +/- 0.4
Airmass coefficient 1 (a1)	1.297 +/- 0.014
Airmass coefficient 2 (a2)	-0.245 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	4.11
Optimal Annulus	8.54
Transit Duration (day)	0.0618 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.013 vs 0.16309 (deviation 2.28σ)
Duration fit vs published	0.0618 d vs 0.0567 d (deviation 1.28σ)
Mid-time O–C	-3.9 min
Depth SNR	13.6
Residual scatter	1.09 %

Light curve

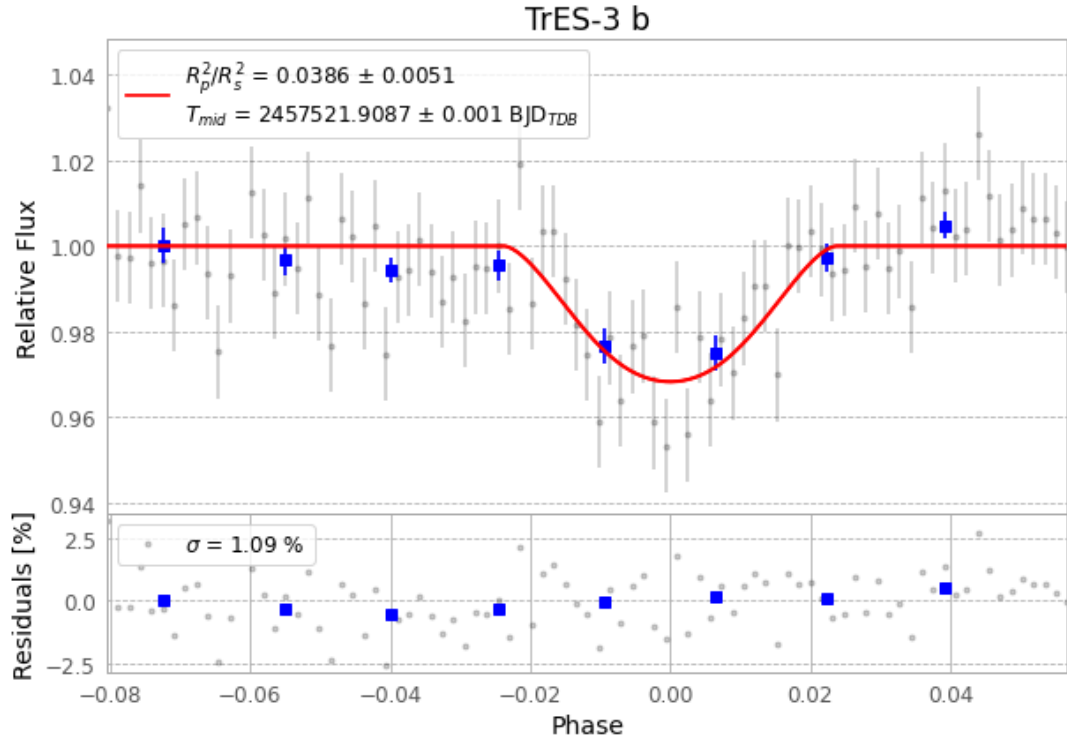


Figure 1 — FinalLightCurve_TrES-3 b_2016-05-13.png (SHA-256 de9adf3954ade13b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [211, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3704f4200b4007f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), TRES-3160513071212.FITS → TRES-3160513113012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160513.log (e8a3c7650e8b...), FinalLightCurve_TrES-3 b_2016-05-13.png (de9adf3954ad...), FinalLightCurve_TrES-3 b_2016-05-13.csv (4d42412ae3e2...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 15:55Z.